

University Number: _____

Seat Number: _____

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data Mining
(Subclass B)

Date: May 13th 2024

Time: 6.30pm-8.30pm

There are 4 questions in total giving the following number of points, respectively: 20, 20, 30, 30. Answer ALL questions. They are all COMPULSORY.

Write your answers on this examination paper in the corresponding space after the questions.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Type of Calculator: _____

Question 1 (20/100). Consider the directed graph G depicted in Figure 1, which is going to be the input for the PageRank algorithm, without introducing random jumps.

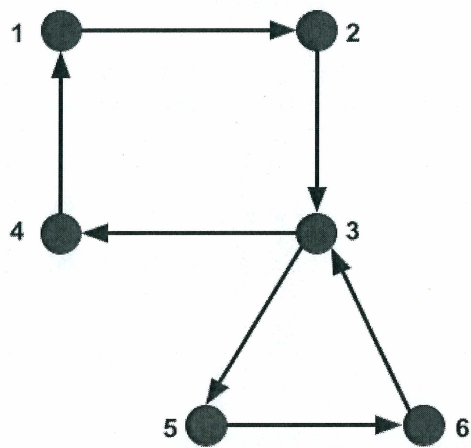


Figure 1. A directed graph G , input for the PageRank algorithm

1.1 (10/100) Compute the PageRank vector of G , that is, the vector that the PageRank algorithm converges to when G is provided in input. Report the PageRank vector below _____

1.2 (10/100) Report the first two vectors obtained as a result of the PageRank algorithm with input G starting from the vector $(1/6, 1/6, 1/6, 1/6, 1/6, 1/6)$

Question 2. (20/100) Consider the training data in Table 1, where there are 3 binary features: 'VE' (Volcanic eruption), 'ICO2' (increase in CO2 emissions), 'ISA' (increase in solar activity), and the binary class 'IT' (increase in average worldwide temperatures). The feature 'Year' only specifies the name of the example and it is not relevant for classification purposes.

Year	VE	ICO2	ISA	IT
1500	Y	N	N	N
1600	N	N	Y	N
1700	Y	N	Y	Y
1800	N	Y	N	N
1900	Y	Y	Y	Y
2000	N	Y	N	Y
2050	N	Y	N	Y
2100	Y	Y	Y	Y

Table 1. Training data for our Naive Bayes classifier.

2.1 (6/100) Predict the class value for (VE='N', ICO2='Y', ISA='N') using the Naive Bayes classifier we presented in our course. _____

2.2 (7/100) What are the two quantities you compared against in order to make such a prediction? Report them as a fraction (e.g. 3/25). _____

2.3 (7/100) Is it necessary to compute Prob (VE='N', ICO2='Y', ISA='N') to make such a prediction using the Naive Bayes classifier? In the positive case, argue why it is important to compute such a quantity. Otherwise, argue why it is important not to compute such a quantity.

Question 3 (30/100). Recall the following clustering algorithm that we called k-means-- in our course. The first centroid is chosen uniformly at random from the set X of input points. For every $1 < i \leq k$, let C be the set of centroids chosen up to step $i-1$. Let x in X be a point that maximizes $d(x, C)$ where $d(x, C) = \min_{y \in C} d(x, y)$. Then $C = C \cup \{x\}$. Those steps are iterated until k centroids are selected, that is $|C|=k$ (we assume $|X| \geq k$). k-means-- might select outliers as centroids. Consider the following algorithm that we call (k,z)-CO. Given a non-negative real number r we denote $S(x, r) = \{y \in X : d(x, y) \leq r\}$, that is the set of points in X that are at distance at most r from x . Given an integer $z > 0$ and a non-negative real number r , (k,z)-CO picks the first centroid to be a point x in X maximizing $|S(x, r)|$. If there are multiple choices, then one of them is selected arbitrarily. Then, it removes $S(x, r)$ from X and iterates those steps until k centroids have been chosen (we assume $|X| \geq k$). If at the end of the algorithm there are some points left in X , those points are called *uncovered*. (k,r)-CO computes the smallest value for r such that there are at most z uncovered points and then it outputs the set of centroids corresponding to that value of r , which are computed by running the steps above.

Consider the following points in the 1D Euclidean space: $P_1=0, P_2=0.2, P_3=0.3, P_4=0.4, P_5=0.8, P_6=1, P_7=1.1, P_8=1.2, P_9=1.3, P_{10}=1.4, P_{11}=3$.

3.1 (10/100) Which point is selected with probability 1 to be a centroid by k-means--?

3.2 (10/100) What is the value for r computed by (k,z)-CO when $k=2$ and $z=2$?

3.3 (10/100) Which points are left uncovered using the value for r specified in 3.2?

Question 4. (30/100) Consider the following variant (that we call H) of the FuDYADT algorithm that we saw in our course. Let ϵ in $(0,1)$. Let S be a set of examples, let $g(S, i)$ be the Gini index of the i th feature on the set of examples S . Let $opt(S)$ be a feature with minimum Gini index in S . Suppose there is a node v in a decision tree T splitting on the i th feature such that

$$g(S(T_v), i) - g(S(T_v), opt(S(T_v))) > \epsilon,$$

where $S(T_v)$ denotes the set of examples associated with T_v . In that case we say that T_v is *damaged*. As examples are inserted and deleted, H rebuilds from scratch any damaged subtree T_v as soon as there is any. In that case, the tree is rebuilt on the *current* set of examples associated with T_v while using v as root. If at any step, there are several subtrees that are damaged, H rebuilds the damaged subtree associated with the largest set of examples. When rebuilding a subtree, H uses the static algorithm that selects the splitting feature to minimize the Gini index. If two splitting features have the same Gini index, the one with the smallest index is chosen. As a stopping condition, $\max \text{height}=2$ (height of root is 0). Consider the training data represented in Table 2, where f_1, f_2, f_3 are binary features and c is the binary class.

	f_1	f_2	f_3	c
x_1	0	0	1	0
x_2	0	0	1	0
x_3	0	0	1	0
x_4	1	0	0	0
x_5	0	1	1	1
x_6	1	0	1	1
x_7	1	0	1	1
x_8	1	0	1	1

Table 2. Set of examples used for training our decision tree.

4.1 (3/100) Build a decision tree with maximum height = 2 on the set of examples in Table 2, using the static algorithm for building a decision tree, which at each step selects the splitting feature to be a feature minimizing the Gini index. Show the tree you built:

4.2 Suppose we remove from the training data in Table 2, the examples x_1 and x_2 , in that order. Use FuDYADT with $\text{eps}=0.49$ to update the decision tree built in question 3.1 after every single update. Recall that FuDYADT only counts how many examples are inserted or deleted when deciding whether to rebuild a tree or not.

4.2.1 (3/100) Is there any damaged subtree after removing x_1 ?

4.2.2 (3/100) Is there any damaged subtree after removing x_2 ?

4.3 Consider again the examples in Table 2 (that is, including x_1 and x_2). Suppose we insert the following examples x_9 and x_{10} , in that order

	f_1	f_1	f_1	c
x_9	1	1	0	1
x_{10}	1	0	0	1

Update the decision tree built in question 3.1 after inserting x_9 and x_{10} by running the H algorithm with $\text{eps} = 0.21$.

4.3.1 (3/100) Is there any damaged subtree after inserting x_9 ? _____

4.3.2 (3/100) Is there any damaged subtree after inserting x_{10} ? _____

4.3.3 (3/100) Show the tree built by H after inserting x_{10}

4.4 Suppose we insert m examples $x_{11}, \dots, x_{10+m}$ to the examples added so far (that is $x_1, x_2, \dots, x_{10}$) each one being a copy of the following example for every $i=1, \dots, m$.

	f_1	f_1	f_1	c
x_{10+i}	0	1	1	1

4.4.1 (3/100) What is the smallest value m' for m so that the whole tree will be rebuilt by H when $\text{eps}=0.06$? _____

4.4.2 (3/100) Show the final tree built by H after those m' examples have been inserted.

4.5 (3/100) What is the most important advantage of FuDYADT when compared against H ?

4.6 (3/100) What is the most important advantage of H when compared against FuFYADT?
